

Depositor Discipline in Community Banking

Evidence from the CECL Information Shock

Dimuthu Ratnadiwakara

Federal Reserve Bank of Richmond

25th FDIC Bank Research Conference

The views expressed are those of the author and do not reflect those of the Federal Reserve Bank of Richmond or the Federal Reserve System.

Do uninsured depositors actually discipline banks?

Theory says they should; causal evidence has been elusive

- ▶ Monitoring by uninsured claimants is the core private check on bank risk-taking
Calomiris–Kahn '91; Diamond–Rajan '01
- ▶ Riskier banks pay higher rates, and their uninsured balances shrink — depositors withdraw, or decline to roll over maturing claims . . .
but is that **discipline** (depositors reacting to news about risk) or **equilibrium sorting** (risky banks and yield-chasing depositors matching)?
- ▶ **Policy stakes are current:** Main Street Depositor Protection Act would insure \$10M per business account; reciprocal networks, sweeps, and brokers already convert uninsured money into insured money Huberdeau-Reid–Pennacchi '25

⇒ **Needed: a shock to information about bank risk that is unrelated to concurrent bank fundamentals**

Why the causal evidence has been elusive: three problems

Why prior estimates conflate information responses with equilibrium sorting

- ▶ **Simultaneity.** A bank's risk profile, deposit mix, and depositor behavior co-evolve — regressions of flows on risk measures capture sorting, not a disciplinary response
- ▶ **Accounting opacity.** Under the incurred-loss standard banks could delay recognition and understate risk in disclosures, so measured risk was partly a reporting choice Beatty–Liao '11; Bushman–Williams '15
- ▶ **Aggregate stress.** Episodes with the most variation in bank distress are exactly those where expanded guarantees and system-wide shocks mute or contaminate the depositor response Berger–Turk-Ariss '15; Bennett–Hwa–Kwast '15

⇒ I argue that the mandatory CECL adoption of 2023 addresses all three

What is CECL? Lifetime expected losses replace incurred losses

ASU 2016-13 (FASB 2016), codified in ASC 326 — the largest change to bank loan accounting in decades

Before: incurred loss (ILM)

recognize a loss only when *probable and estimable*
⇒ provisions lag defaults, disclosures understate credit risk

After: expected loss (CECL)

reserve for *lifetime* expected credit losses at origination and every reporting date ⇒ credit risk forced into a reported number

At transition, the entire existing loan book is re-reserved on a single day:

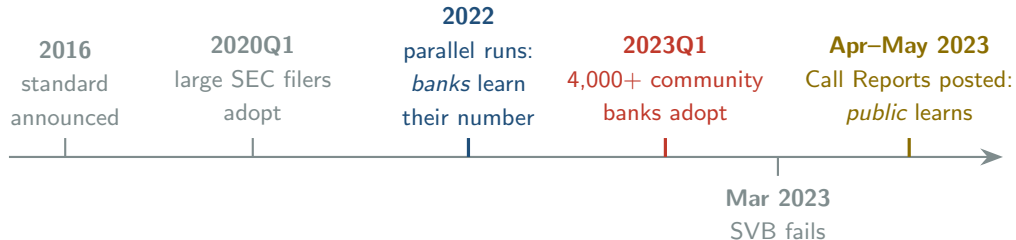
$$\Delta_i = ALLL_{i,t^*}^{CECL} - ALLL_{i,t^*-1}^{ILM} \quad (\text{one-time charge to retained earnings, no cash flow})$$

$$\text{Treatment:} \quad CECL \text{ Adj}_i = \frac{\Delta_i}{Equity_{i,t^*-1}} \times 100$$

⇒ Δ_i reflects the composition and seasoning of loans made years earlier — not deposit-market conditions in 2023

The experiment: a mandatory disclosure with a predetermined size

CECL Day-One adjustment = one-time charge to retained earnings, fixed by past underwriting



- ▶ Pure accounting event: no cash flow, no change in loan risk — only **news** about lifetime expected losses already embedded in the book
- ▶ Everyone knew the date; nobody outside the bank knew the number

Back to the three problems: what this shock delivers

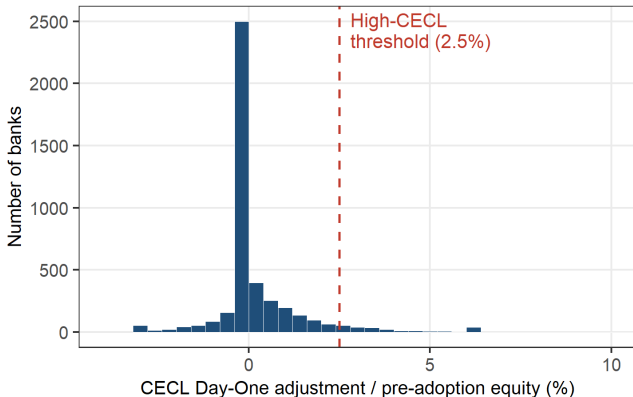
Each identification challenge, and the feature of the Day-One adjustment that addresses it

Problem	Feature of the CECL Day-One charge
Simultaneity	Size is predetermined : set by the composition and seasoning of loans underwritten years earlier; 2022Q4 observables explain $\sim 1\%$ of its variation
Accounting opacity	Mandatory and standardized : one rule, every bank, one date, disclosed on Schedule RI-A — no discretion over whether to reveal
Aggregate stress	Cross-sectional intensity within one quarter — calendar-quarter FE absorb the tightening cycle and the 2023 stress; bank $\times$ quarter FE absorb them bank by bank

⇒ **And a timing wedge: banks knew their number in 2022, the public only in April–May 2023**

Treatment: the disclosed charge, scaled by equity

4,320 community banks (<\$10B); charge disclosed on Schedule RI-A of the adoption-quarter Call Report



92% adopt in 2023Q1

~**half** disclose a charge of about zero

217 High-CECL banks: charge $\geq 2.5\%$ of equity (≈ 95 th pctile), mean 4% of equity

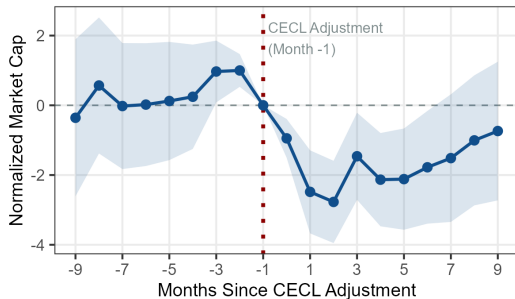
Charge is hard to predict: 2022Q4 observables explain $\sim 1\%$ of its variation

[▶ detail](#)

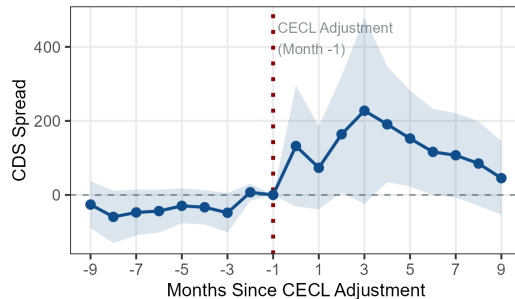
The Day-One number is news: markets repriced it in 2020

Large SEC filers adopted 2020Q1, three years before community banks; monthly event studies, coefficient on charge/equity

Market capitalization



5-year CDS spread



⇒ Equity falls $\sim 2.8\%$ per pp of charge at adoption; CDS spreads widen with a lag — sophisticated investors treat the number as material

Data: a decade of quarterly Call Reports for 4,320 community banks

FFIEC 041/051, 2016Q1–2025Q4 (~97,000 bank-quarters); RateWatch posted rates; CRSP + Bloomberg for the 2020 cohort

- ▶ Federally insured banks < \$10B in assets at 2022Q4; median bank \$0.3B
- ▶ Deposit breakdowns by insurance status from Schedule RC-E; outcomes scaled by assets
- ▶ Time deposits fund **12.7%** of assets (5.3 uninsured + 7.4 insured)

Pre-adoption means	High CECL	Low CECL	Hi–Lo
CECL Adj / Equity (%)	3.99	0.08	3.91***
NPL / Loans (%)	0.60	0.45	0.15***
Equity / Assets (%)	8.88	10.34	–1.46***
Ins. time dep. / Assets (%)	9.13	7.27	1.86***
Unins. time dep. / Assets (%)	5.62	5.28	0.33

High-CECL banks: weaker loan quality, thinner capital — but **balanced on the primary outcome** (bank FE absorb the levels)

Empirical strategy: two specifications

Bank + calendar-quarter FE; SEs clustered by bank; controls: lagged log assets, equity/assets, int. exp./assets

Specification 1 — level DiD across banks (expected sign $\beta < 0$)

$$y_{it} = \alpha_i + \gamma_t + \beta \text{Post}_{it} \times \text{CECL}_i + X'_{it-1} \delta + \varepsilon_{it}$$

CECL_i : continuous CECL Adj or High-CECL indicator; event studies replace Post with event-time dummies, $k = -1$ omitted. Quarter FE absorb the tightening cycle and any aggregate 2023-stress response.

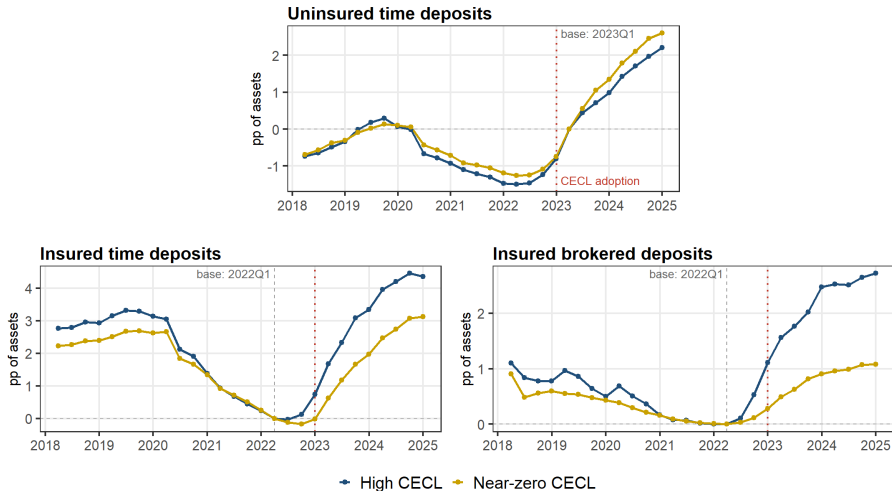
Specification 2 — triple difference within bank $\times$ quarter (series stacked)

$$y_{idt} = \alpha_{it} + \beta_2 \text{Post}_t \times \text{Unins}_d + \beta_3 \text{CECL}_i \times \text{Unins}_d \\ + \beta_4 \text{CECL}_i \times \text{Post}_t \times \text{Unins}_d + \varepsilon_{idt}$$

α_{it} absorbs every bank-level shock in every quarter — funding demand, stress exposure, liability strategy; β_4 is identified only from the within-bank uninsured-vs-insured differential. Permutation inference (1,000 reassignments): triple diff $p = 0.001$.

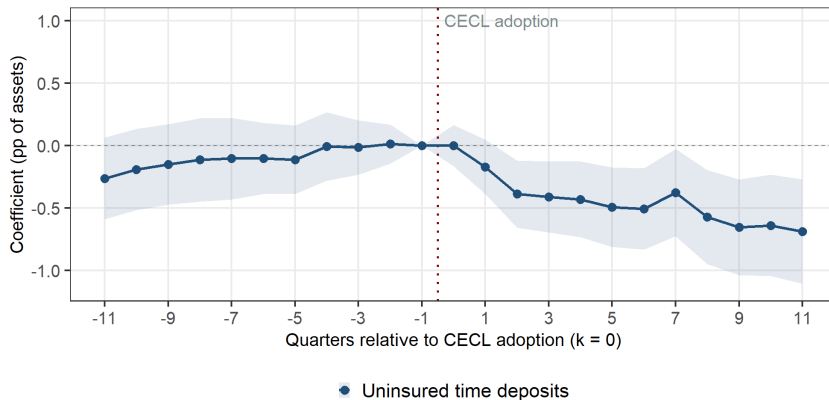
Before any regression: uninsured money leaves, insured money arrives

Group means, pp of assets; bases 2023Q1 (top) and 2022Q1 (bottom); 206 vs 2,265 banks; no controls or FE



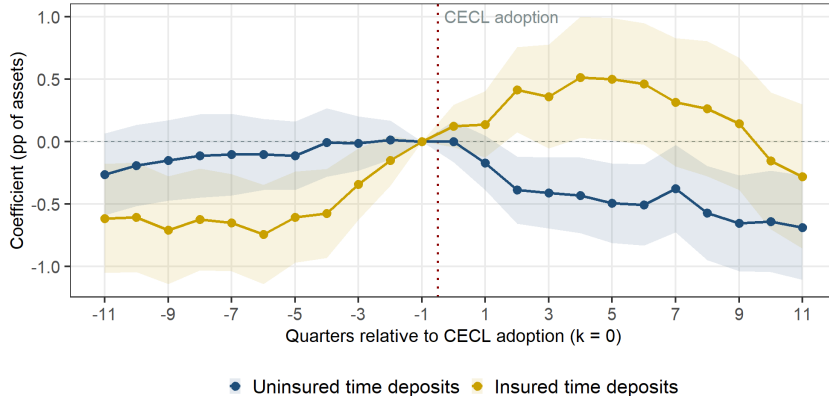
Uninsured time deposits are flat for ten quarters... then break at disclosure

Event study, High-CECL $\times$ event time; bank + quarter FE; 90% CI



...and insured time deposits rise to meet them

Same specification, insured series overlaid; identical axes



Magnitudes: a composition shift, not a funding collapse

Coefficients in pp of assets; bank-clustered inference; */**/** = 10/5/1%

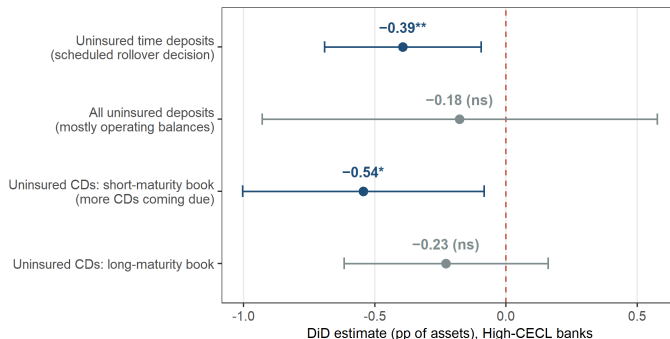
► specification

	Unins. time dep. / assets	Ins. time dep. / assets	Unins. – ins. gap (composition)
Post × High CECL	−0.39**	+0.71**	−1.25***
Post × CECL Adj (continuous)	−0.072**	+0.070	−0.177***
Bank + quarter FE	✓	✓	—
Bank × quarter FE	—	—	✓
Observations	97,064	97,064	194,128

⇒ The gap survives absorbing every bank-level shock in every quarter — stress cannot flip uninsured and insured CDs in opposite directions inside the same bank

Discipline lives exactly where the depositor holds a decision

Same specification as the table — interaction coefficients shown graphically with 90% CIs from here on

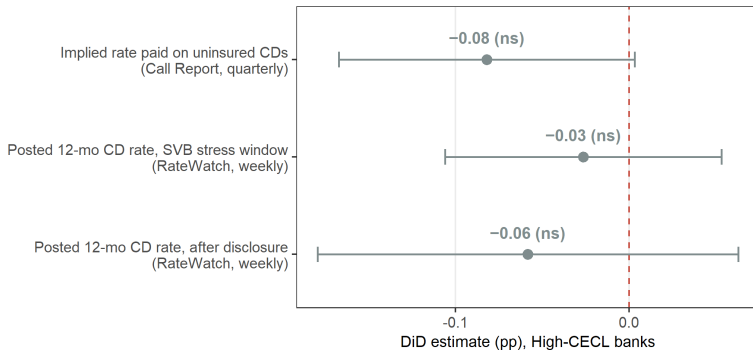


⇒ **No response in operating balances; twice the response where more CDs came due — a quantity response at the rollover margin**

High-CECL banks never defended their funding on price

Implied rates from Call Report interest expense; posted APYs from RateWatch, bank-week panel

[▶ detail](#)

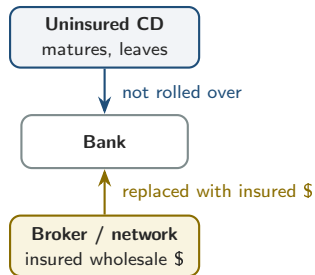
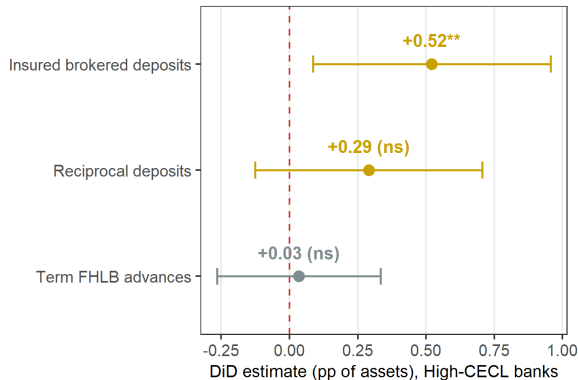


⇒ **Three nulls, two datasets: no repricing in the stress weeks, none after disclosure — exactly what the rate-setting literature predicts** Begenau–Stafford '23; Egan

et al. '25

Where the money went: banks repurchased the lost insurance

DiD estimates for the three replacement margins, High-CECL banks, pp of assets

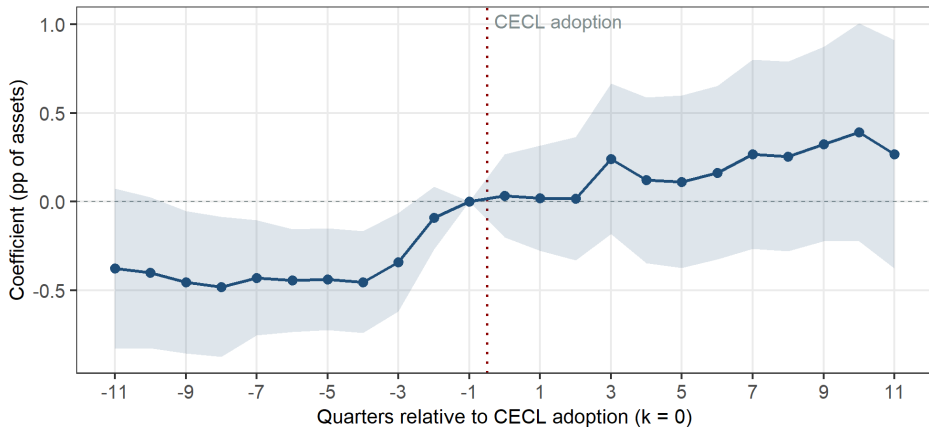


Balance sheet keeps its size — only the *insurance status* of the funding moves. No move in uninsured brokered money or FHLB borrowing.

Two actors, two clocks: banks moved first, on private information

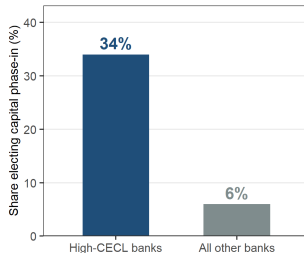
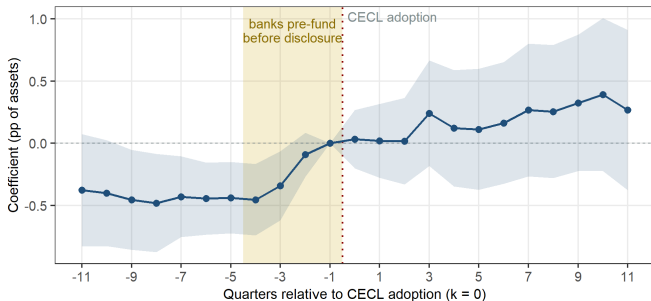
Insured brokered deposits, event study; High-CECL $\times$ event time; 90% CI

► phase-in mechanics



Two actors, two clocks: banks moved first, on private information

Banks ran CECL in parallel through 2022 — management knew the number before the public did



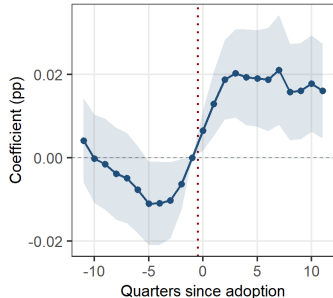
Pre-adoption ramp scales with the *still-undisclosed* charge (0.53***); banks electing capital relief pre-funded 0.41** more

⇒ Bank margins move on private information, before adoption; depositor margins move only when the number becomes public

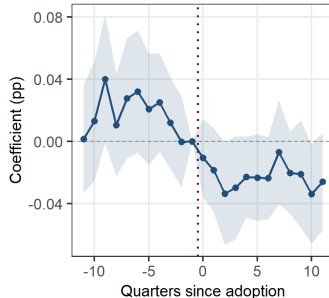
The substitution is costly: funding costs up, profitability down

Event studies for interest expense, ROA, NIM; High-CECL $\times$ event time; 90% CI

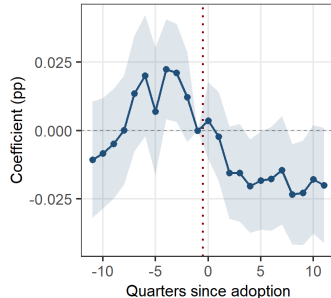
Interest expense $\uparrow$ 8–9bp/yr



ROA $\downarrow$ 15bp/yr



NIM $\downarrow$ 9bp/yr

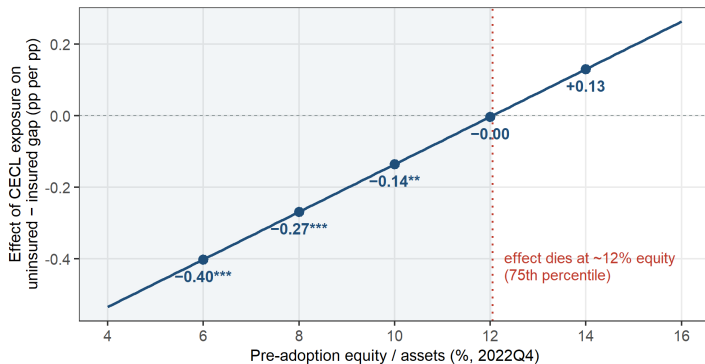


$\Rightarrow$ They paid no premium to stayers and posted no higher rates — the cost is the substitution: maturing uninsured CDs replaced with insured wholesale money raised at wholesale rates

The response scales with distance to default

DiD interacted with pre-adoption (2022Q4) equity/assets; gap outcome; effect of the continuous charge

► full spec

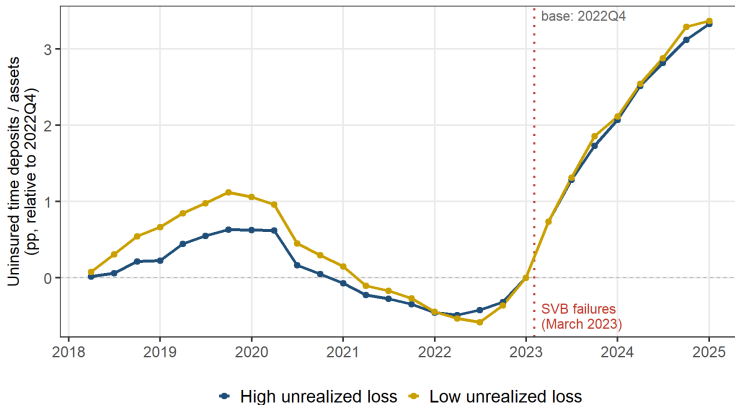


⇒ A given charge is worse news at a thin buffer — monitoring intensity tracks what the disclosure implies about distance to default

Placebo: the same picture, drawn on stress exposure, is flat

Zero-charge banks split at the median 2022Q4 unrealized loss; group means, no controls or FE

▶ placebo detail

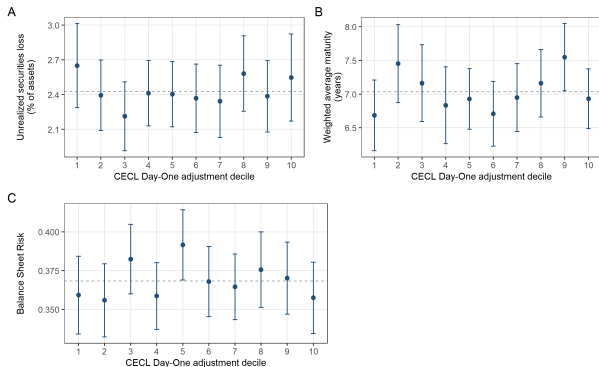


⇒ Banks with no CECL news show no separation on run exposure — 0.04pp at end-2024, against 0.40pp for the disclosed charge

This is not the SVB stress in disguise

2022Q4 stress-vulnerability characteristics across deciles of the disclosed charge (positive-adjustment banks)

► placebo detail



- Treatment $\perp$ run exposure: unrealized losses, asset maturity, Balance Sheet Risk Huberdeau-Reid-Pennacchi '25 all flat; <1% of variation explained
- **Placebo:** at zero-charge banks, uninsured CDs did *not* fall with unrealized losses after March 2023 (wrong sign: +0.07)
- Horse race: CECL estimates unchanged
- 2023 runs were wholesale, at public banks; this sample is private and retail-funded Cipriani et al. '24

What I cannot rule out: attention

The design identifies the response to the disclosure, not how much attention was already in the room

- ▶ The tests above rule out that *treatment* is stress exposure: the charge is orthogonal to run vulnerability, the placebo on unrealized losses is flat, and posted rates show no differential stress response
- ▶ They do *not* rule out that 2023 was a period of unusually high attention to bank fragility. Depositors who had just watched two banks fail may read a credit-risk disclosure more carefully than they otherwise would
- ▶ If so, the estimates are the response *conditional on* attention, not an unconditional response to disclosure — an upper bound on what the same number would do in a quiet year
- ▶ That is an external-validity limit, not a threat to identification: heightened attention applies to treated and control banks alike, and cannot manufacture a gap that lines up with a predetermined accounting charge
- ▶ Testing it needs a disclosure shock in a calm period. I do not have one here

Takeaways

A mandatory credit-risk disclosure moved uninsured funding at solvent community banks

1. **Depositors read the number and acted.** Uninsured CDs fell and insured CDs rose in proportion to the disclosed charge — flat for five years before, breaking only at public disclosure
2. **Discipline = quantity exit at the rollover margin.** No repricing, no move in operating balances, stronger where more CDs came due and where capital buffers were thin
3. **Banks repurchased the insurance** — insured brokered money replaced monitored money; interest expense rose and profitability fell: the bite of discipline is a price, paid by the bank
4. **Private coverage may already be dampening discipline.** Reciprocal networks, sweeps and insured brokered CDs convert uninsured balances into insured ones with no change in the balance sheet — the monitored share of funding shrinks before any statutory change
5. **Policy:** expanding coverage (e.g. \$10M business accounts) converts exactly the funding that monitors into funding that does not — weigh the discipline displaced against the stability gained

Backup: the charge is hard to predict

Cross-section of the Day-One charge on 2022Q4 observables

- ▶ Loads modestly on NPL ratio, size, capital, deteriorating ROA trend — directions consistent with credit-sensitive books needing larger lifetime reserves
- ▶ Adjusted $R^2 \approx 1.4\%$ in the most saturated specification (fundamentals + deposit mix + performance trends + local market HHI)
- ▶ High-CECL banks are *not* observably weak banks relabeled: almost all variation is orthogonal to anything a depositor could see before disclosure

Backup: rate tests in detail

Implied rates: Call Report interest expense on time deposits above/below \$250k (2017–)

- ▶ Implied rate on uninsured CDs, Post $\times$ High-CECL: -0.082 (SE 0.052) — no premium extracted by depositors who stayed
- ▶ Within-bank large–small rate spread: -0.110^* — if anything narrows
- ▶ RateWatch posted APYs (weekly, $\sim 3,300$ banks): 12-mo CD \$10k/\$100k tiers and savings, no differential move in stress window (-0.026 , SE 0.048) or post-publication (-0.058 , SE 0.074)
- ▶ $>90\%$ of branches follow centrally set rates Begenau–Stafford '23; healthy banks can bid for uninsured money, weak banks cannot Maechler–McDill '06

Backup: the capital phase-in election

Call Report Schedule RC-R Part I item 2.a; elected on the first CECL filing, irrevocable

- ▶ Election spreads the Day-One hit to *regulatory* capital over three years: 75% added back in year one, 50%, 25%, full impact from year four
- ▶ Economically sensible only if the bank expects a material hit — election probability rises 0.072*** per pp of charge
- ▶ 34% of High-CECL banks elect vs. 6% of the rest; electors' pre-adoption insured-brokered ramp is 0.41** pp larger
- ▶ Election is measured *at* the first CECL filing; only the brokered ramp is genuinely quarters early

Backup: capital-buffer interaction, full specification

Moderator: equity/assets frozen at 2022Q4, winsorized 1/99; gap outcome = uninsured – insured time deposits/assets

- ▶ $\text{Post} \times \text{CECL Adj} \times \text{Equity}_{22Q4}$: $+0.0665^{***}$ (0.019); binary analog $+0.31^{***}$
- ▶ Main effect at zero equity: -0.80^{***} (continuous), -4.12^{***} (binary)
- ▶ Level-outcome analog: $+0.0275^{**}$; robust to dropping the time-varying equity control
- ▶ Implied effect crosses zero at equity $\approx 12\text{--}13\%$, just above the 75th percentile — best-capitalized quartile shows essentially no response

Backup: SVB placebo and horse race

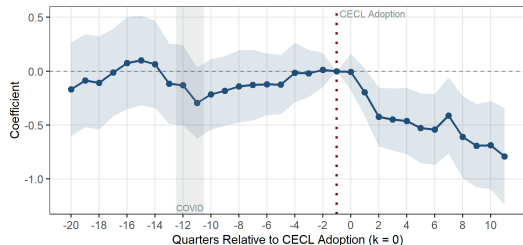
DiD around March 2023 with 2022Q4 unrealized securities loss as intensity

- ▶ Zero-charge banks ($\sim 2,278$, no CECL news): $\text{Post-SVB} \times \text{loss} = +0.073^{**}$ for uninsured CDs — *wrong sign* for a run; insured CDs move the same way ($+0.062^*$) $\Rightarrow$ rate-cycle funding mix, not flight
- ▶ Full-sample horse race: CECL coefficients essentially unchanged (-0.078^{**} continuous, -0.406^{**} High-CECL); loss interaction small and insignificant
- ▶ Uninsured deposit share enters the charge cross-section *negatively* (-0.003^{**}): high-CECL banks held slightly fewer of the deposits that ran in 2023

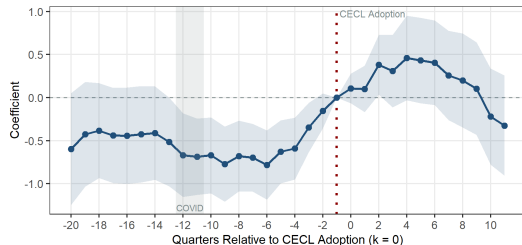
Backup: five-year pre-period, balanced endpoints

Extended event studies from 2018Q1; edge bins pool $k \leq -20$ and $k \geq +11$; joint pre-trend tests

Uninsured time deposits ($p = 0.18$)



Insured time deposits ($p = 0.37$)



Late upward drift in the insured series is the banks' own pre-funding footprint (insured brokered CDs are booked as time deposits); absorbed into the $k = -1$ baseline, so post coefficients *understate* the total movement.